AI For Literature Reviews

CHEM-CIVE-ENVR 5670
Professor Stubbins
Fall 2026

Northeastern University
NULab for Digital Humanities and Computational Social Science

Feel free to ask questions at any point during the presentation!

Discussion

- Has anyone ever used ChatGPT or another generative AI?
- What do you think it did well or poorly?
- Were there any aspects of the responses you found odd or misleading?

Workshop Agenda

- Generative AI
  - AI and ‘Truth’-Confabulation/Hallucination
- How to use ChatGPT and Claude for Literature Reviews (Demonstration and Exercise)
- Ethics in AI
  - Bias in AI and Countering Bias
  - Personal Impacts
- Other AI-Powered Literature Review Tools
- Conclusion

Slides, handouts, and data available at

https://bit.ly/fa26-stubbins-AIliteraturereview

Feel free to ask questions at any point during the presentation!

Generative AI

Important questions

- How do human biases impact generative AI model outputs?
- How can we counter the weaknesses of current AI models?
- How can we integrate generative AI with other tools and practices?

Vocabulary

- Artificial Intelligence (AI): A technology that "learns" from datasets to solve problems and mimic human intelligence.
- Large Language Model (LLM): Powerful AI systems designed to understand, generate, and manipulate human language. E.g. GPT4
- Generative AI: It uses AI algorithms that are trained on big datasets in order to create new content, including text, images and audio, in response to a query or prompt. E.g. ChatGPT, Claude, DALLE, Midjourny.
- Bias in AI: The biased outputs due to human biases that existed in original training data or skew the AI algorithm. Could result in potential harms.

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Generative AI: Introduction

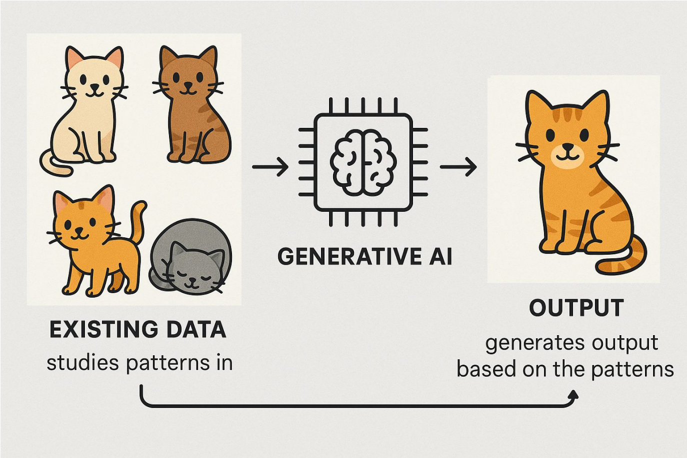
AIs like ChatGPT and Claude Sonnet 4.6 (which generate text) and DALL-E and Midjourney (which generate images) work by training on very large datasets of information and then mimicking that training data to produce "new" outputs.

Feel free to ask questions at any point during the presentation!

Generative AI: How it Works

- Generative AIs are fundamentally based on statistics. A text-generation AI like ChatGPT generates words according to patterns learned from the training data.
- Similarly, an image-generation AI learns patterns from existing data and generates images that “looks right” but do not exist in the data.

Generated by GPT4o
Aug 15th 2025



Example Prompt: Claude Sonnet 4.6 (1/2)

Prompt: In a few sentences, explain Earth Systems Thinking. (Claude Sonnet 4.6, Anthropic, Accessed June 7th, 2026)

Response: Earth Systems Thinking is a framework that views Earth as a collection of deeply interconnected subsystems — the atmosphere, hydrosphere, biosphere, lithosphere, and others — that constantly interact and influence one another. Rather than studying these components in isolation, it emphasizes understanding the feedback loops, cycles, and emergent behaviors that arise from their interactions (for example, how ocean temperatures affect atmospheric circulation, which in turn shapes precipitation and vegetation patterns). **(Continued on next slide)**

Example Prompt: Claude Sonnet 4.6 (2/2)

Prompt: In a few sentences, explain Earth Systems Thinking.

Response: (Continued from previous slide) It recognizes that changes in one part of the system can cascade in complex, sometimes non-linear ways through the whole, making prediction and management of environmental issues far more nuanced than simple cause-and-effect thinking would suggest.

How the Earth Systems Paragraph Was Generated

- A large language model analyzed writing on Earth Systems Thinking and identified common patterns
- These patterns are recreated in the generated text
  - The response uses keywords related to Earth Systems Thinking (atmosphere, hydrosphere, biosphere, etc.) and a tone common to scientific writing
  - The argument about interconnectedness is also a repeating pattern in writing on Earth Systems Thinking

Feel free to ask questions at any point during the presentation!

Exercise: Prompting Claude

Try asking Claude to explain a topic you are familiar with and consider the response:

- What did it get right?
- What did it get wrong?
    - Why do you think it made those mistakes?
- Did it miss anything important?

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AI And ‘Truth’

Confabulation/Hallucination

- Confabulation refers to the phenomenon where large language models (LLMs) generate incorrect, nonsensical, or fabricated information, even when presented with seemingly clear and accurate prompts.
- AI confabulation is also referred to as hallucination.

Generative AI and "Truth"

- Text-generation AIs aim to produce text that is grammatically correct and linguistically probable.
  - They do not understand "facts," only patterns of word use.
- They can generate truthful text, but also frequently create confabulations/hallucinations.
  - When asked to generate citations, they can generate plausible-looking but fake sources.
  - They may link real but irrelevant sites as sources for made-up facts. They may also invent URLs that do not work and have never worked.
  - AI responses may depend heavily on the phrasing of the prompt, with the AI offering very different responses to similar questions.

Study: Flaws in AI medical advice (1/3)

- NPR reports (March 2026) that a study in *Nature Medicine* reveals several potential problems when AIs are used for medical advice (more than 40 million people have consulted ChatGPT with medical questions, also according to the article). Problems researchers found include:
  - Incorrect identification of conditions.
  - Incorrect advice about appropriate next steps.

ChatGPT might give you bad medical advice, studies warn

MARCH 11, 2026 · 11:21 AM ET

By Katia Riddle

Study: Flaws in AI Medical Advice (2/3)

- “In one scenario, two different users gave slightly different depictions of the same scenario. One of them described ‘the worst headache I've ever had,’ and was directed by the AI to go to the emergency room immediately. The other—who did not use that explicit description—was told to take aspirin and stay home. ‘Turns out this was actually a life-threatening condition,’ says Bean” (Andrew Bean, one of the study’s authors).

ChatGPT might give you bad medical advice, studies warn

MARCH 11, 2026 · 11:21 AM ET

By Katia Riddle

Feel free to ask questions at any point during the presentation!

Study: Flaws in AI Medical Advice (3/3)

- “"When there was a textbook medical emergency, ChatGPT got it right,” according to Girish Nadkarni, another of the study’s authors, but in “more complicated scenarios in which there was an ‘element of time’ at play—the bot often both over- and under-estimated the amount of time a patient could wait until pursuing care.”

ChatGPT might give you bad medical advice, studies warn

MARCH 11, 2026 · 11:21 AM ET

By Katia Riddle

Truth and AI: Example citation prompt (1/3)

Prompt: "Generate a list of 4-5 academic citations on environmental studies in the Bolivian Amazon."
(Claude Sonnet 5, July 29, 2026)

1. Bottazzi, P., & Dao, H. (2013). *On the road through the Bolivian Amazon: A multi-level land governance analysis of deforestation.* Land Use Policy, 30(1), 137.
2. Study on land cover change and fragmentation in Pando, northern Bolivia (1986–2005).

Page number is actually 137-146.

Links to a real article, but does not provide requested citation.

Truth and AI: Example citation prompt (2/3)

Prompt: “Generate a list of 4-5 academic citations on environmental studies in the Bolivian Amazon.”
(Claude Sonnet 5, July 29, 2026)

3. Deforestation scenarios for the Bolivian lowlands.

Links to a real article, without providing a citation.

4. Environmental problems in the Bolivian Amazon caused by deforestation, contaminated rivers, and penetration into protected areas (2024).

Links to a real article, without providing a citation.

Truth and AI: Example citation prompt (2/3)

Prompt: "Generate a list of 4-5 academic citations on environmental studies in the Bolivian Amazon."
(Claude Sonnet 5, July 29, 2026)

5. Pacheco, P. (1998). *Estilos de Desarrollo, Deforestación y Degradación de los Bosques de las Tierras Bajas de Bolivia.* La Paz: Center for International Forestry Research (CIFOR), Centro de Estudios para el Desarrollo Laboral y Agrario, and Taller de Iniciativas y Estudios Rurales y de Reforma Agraria.

Correct, though Claude does not seem to notice that the source is in Spanish. Citation mixes the two languages.

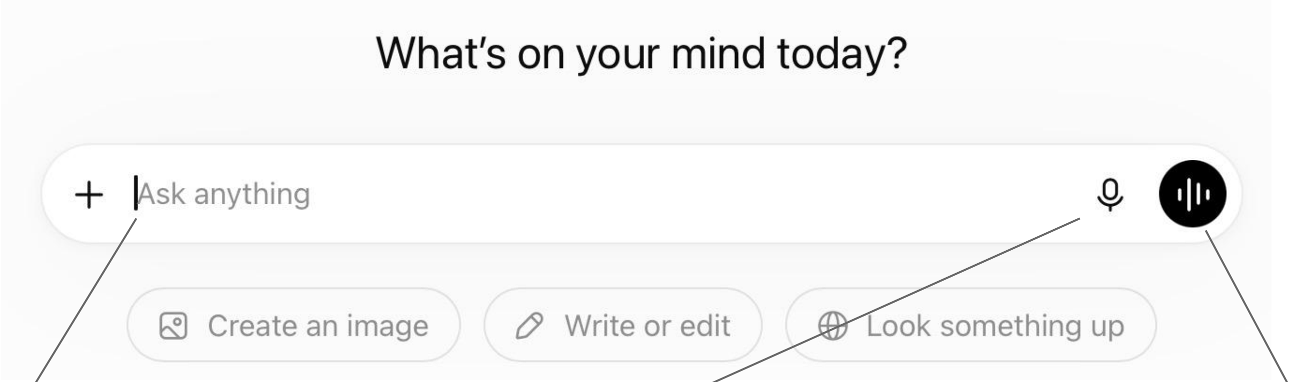
Generative AI and ‘Truth’ summary of literature

- AI may inaccurately attribute concepts to individuals or omit co-contributors and earlier influences.
- It may lack citations or invent fake references.
- It may provide a surface level summary that is over simplified and misleading.
- It may overemphasize sources it has access to, such as those that are not in a paywalled database, and which are often older.

How to use ChatGPT and Claude for literature reviews

Feel free to ask questions at any point during the presentation!

Prompting ChatGPT (1/2)



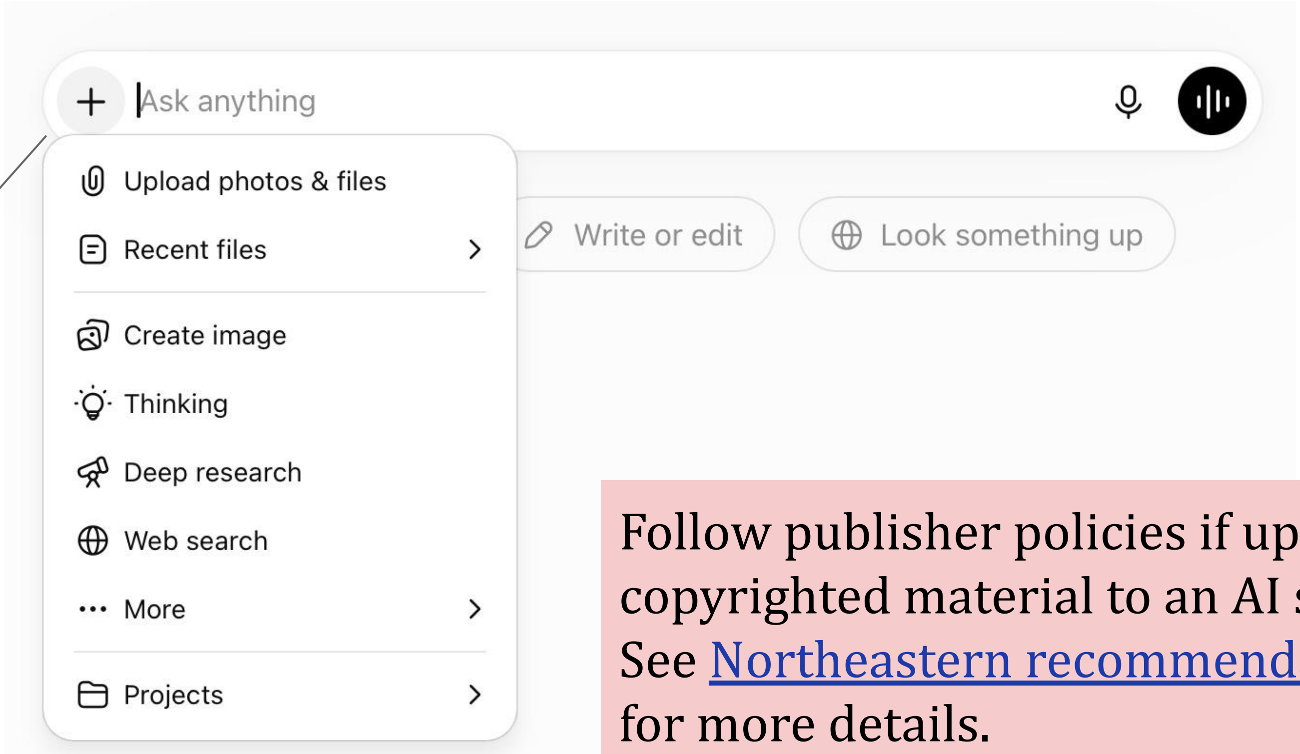
Enter prompt in text format

Enter prompt as speech (transcribed automatically)

Use voice mode

Feel free to ask questions at any point during the presentation!

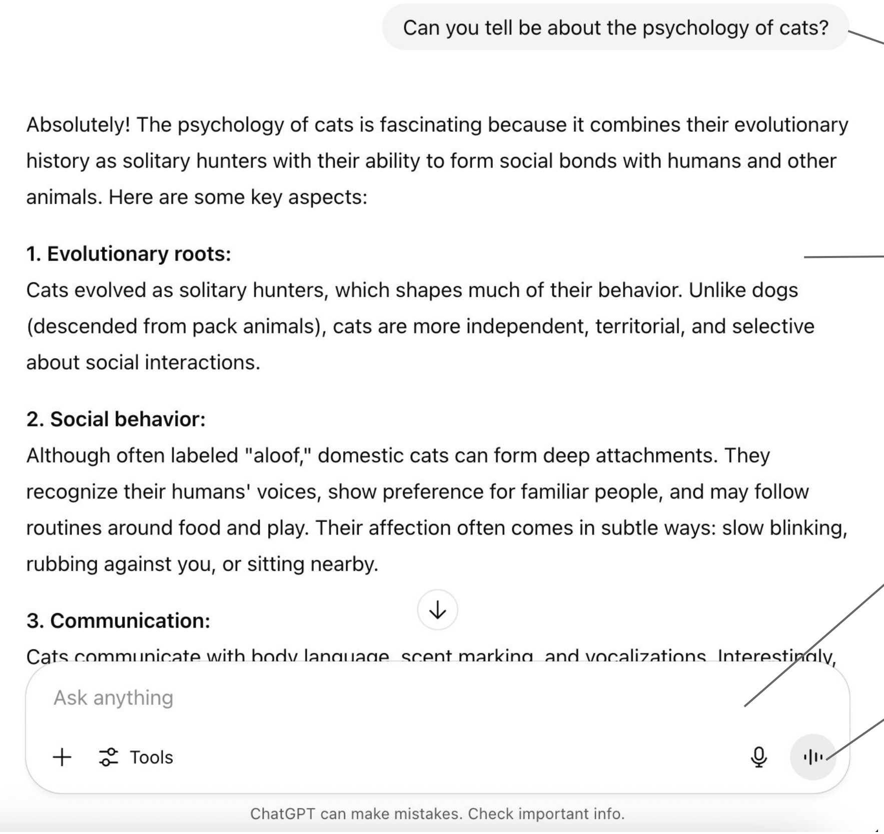
Prompting ChatGPT (2/2)



Upload files and Other tools

Follow publisher policies if uploading copyrighted material to an AI system. See Northeastern recommendations for more details.

ChatGPT Results



Initial prompt

Response

Enter subsequent prompts as text or speech

Use voice mode

Prompting Claude

Follow publisher policies if uploading copyrighted material to an AI system. See [Northeastern recommendations] for more details.

Enter prompt in text format

Voice mode

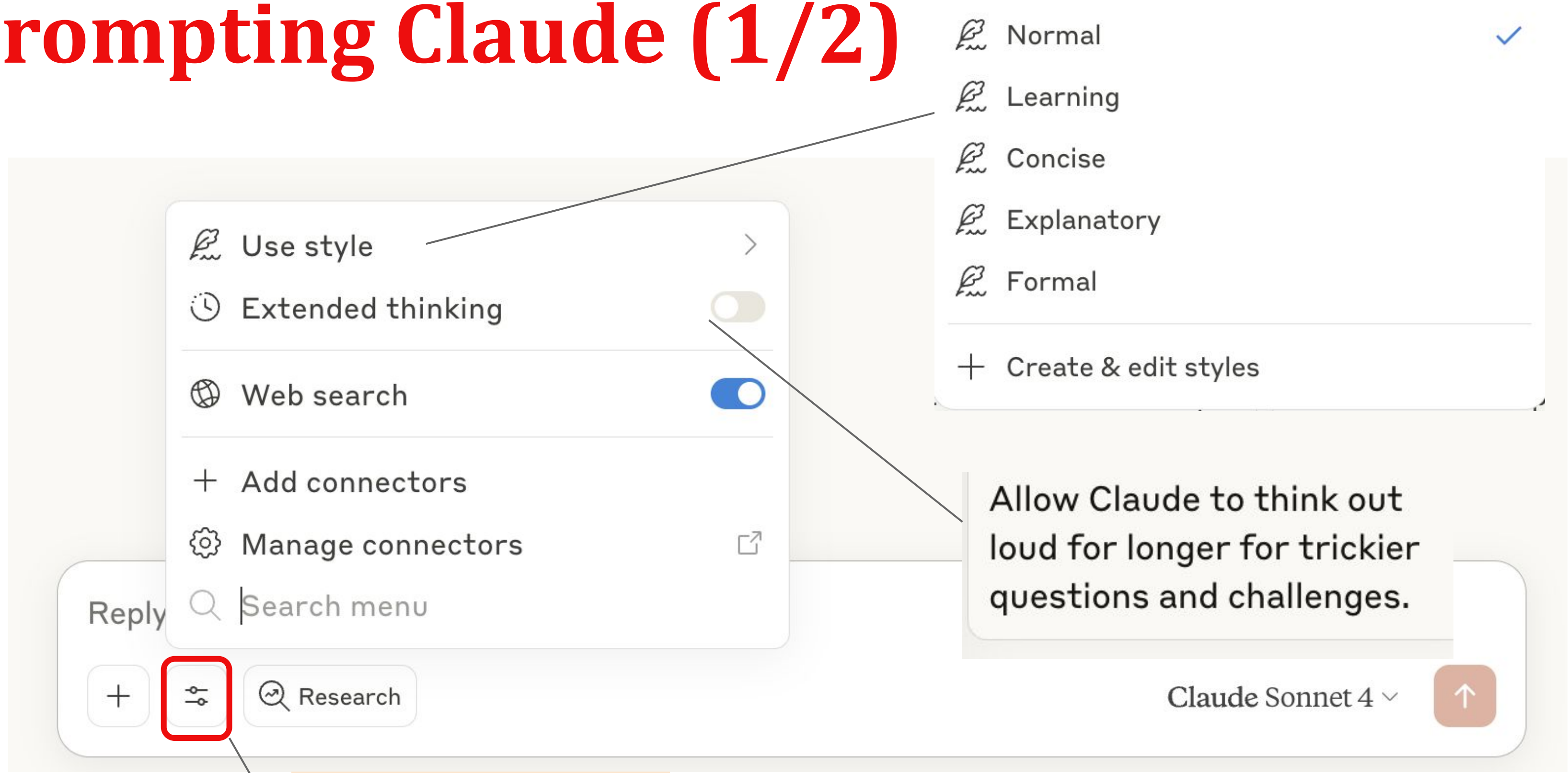
Upload files

Select models

Level of effort

Press and hold to record the prompt

Prompting Claude (1/2)



Open menu

Prompting Claude (2/2)

SM Can you provide a list of common biases in AI generated literature reviews?

Here are common biases that can occur in AI-generated literature reviews:

Selection and Search Biases

- **Training data bias** - AI models reflect the biases present in their training datasets, which may overrepresent certain types of research, languages, or publication venues
- **Recency bias** - Overemphasis on recent publications while undervaluing foundational or older work that remains relevant
- **Publication bias** - Tendency to focus on published studies while missing important unpublished research, gray literature, or negative results
- **Language bias** - Disproportionate emphasis on English-language publications, missing valuable research in other languages

Analytical and Interpretive Biases

- **Confirmation bias** - Selectively highlighting studies that support predetermined conclusions while downplaying contradictory evidence
- **Cherry-picking** - Focusing on specific findings from studies while ignoring conflicting results within the same research

Reply to Claude...

Research

Claude Sonnet 4

Initial prompt

Initial response

Enter subsequent prompts as text

Feel free to ask questions at any point during the presentation!

Your turn

- Go to claude.ai, litmaps, or https://scite-ai.
- Prompt the AI tool you chose to generate a list of sources relevant to your project.
- Identify any issues in the response (and if you identify issues, try another prompt to address them).
- Next, search for your topic in one of the relevant library databases (ex. ScienceDirect).
- Evaluate the strengths and limitations of the literature reviews from each source.

Discussion: Further Inquiry

- Was anything surprising about the responses?
- Did the sources returned by the AI differ from those found in the library databases, if so, how?
- What other types of sources could be useful for checking ChatGPT's and Claude's responses?

Ethics in AI

Feel free to ask questions at any point during the presentation!

Ethics: Generative AI "Originality"

- Some argue that all AI-generated output constitutes plagiarism and copyright infringement, since it is remixing training data that was scraped from the internet without permission from the original creators.
- Many AI companies are facing lawsuits from people whose content was used as training data without their consent.
- Some publication venues, like the *Science* journals, have made it an official policy that AI does not meet the standard for authorship and require authors to disclose use of AI.

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Ethics: Generative AI Training

- AI training data is sometimes supplemented by labels (annotations) added by people (Amironesei and Díaz, 2024). These labels can worsen bias in training datasets.
- People from middle and low income countries often labor in poor working conditions to annotate data for clients in high income countries. Fieldwork by Muldoon et al. (2023) revealed that workers faced traumatizing content, in addition to experiencing discrimination in the workplace and receiving low wages.
- Public awareness can help pressure companies to adopt good practices. For more information on fair labor in AI, see the report AI for Fair Work: From principles to practices by *Fairwork*

Feel free to ask questions at any point during the presentation!

Ethics: Environment (1/2)

- Training and using AI requires processing very large amounts of data, which is done in data centers
- These data centers can have a negative impact on the environment and communities
- Given their intensive energy and water demands, data centers can worsen local water scarcity and increase electricity prices
- To explore how the energy use of AI compares to other digital tasks, check out Jon Ippolito's "What Uses More" app
- For more information on carbon emissions from running computer code (including for AI) check out the Python library CodeCarbon.

Ethics: Environment (2/2)

- A large data center may use 5 million gallons of water per day, equivalent to the water use of a town of about 10,000 to 50,000 people.
- One 100-word AI prompt uses roughly the equivalent of one bottle of water.
- Researchers have come to these figures by calculating the amount of water used in the centers themselves (mostly for cooling), the water used in power plants that supply energy to data centers, and water consumption in the manufacturing of processor chips.

Source: ‘Data Centers and Water Consumption’ by Miguel Yañez-Barnuevo (EESI)

One Note on Ethics and AI

- These slides (and the ones which follow) do not cover all of the ethical considerations of AI development/use. This is a quickly changing field, and there are no complete guidelines or perfect authorities on ethics and AI. Much remains to explore beyond what we are discussing today.
- Starting points for further exploration:
  - Ethics in AI: Why It Matters
  - Recommendation on the Ethics of Artificial Intelligence
  - A Guide to AI Ethics

Bias in AI

What is AI bias?

- Biased outputs arise in AI because of human biases in the original training data, which AI reproduces by recreating patterns found in that original data.
- For example, when AI was used to summarize medical notes, “Google’s AI tool Gemma described men’s health issues with terms like “disabled,” “unable,” and “complex” significantly more often than women’s, who were often framed as more independent despite similar needs, an alarming gender bias trend.” (Source: crescendo.ai)
- What is the ideal “unbiased” scenario? Is an “unbiased” scenario possible?

Bias in AI: Summary

- AI training data reflects the injustices and biases of the society that produced it.
- These biases can be amplified when they are input as training data into an AI, because they seem to be the "right" answers (Dwivedi et al., 2023).

Feel free to ask questions at any point during the presentation!

Bias in AI: Example

Example: "Generate a list of 6-8 citations on the climate tipping point."
(Claude Sonnet 4.6, June 6, 2026)
Reponse:

1. Lenton T.M., Held H. et al. (2008)-Tipping elements in the Earth's climate system.
2. Scheffer, M. Carpenter, S. et al. (2001)- Catastrophic shifts in ecosystems.
3. Kreigler, E. J.W. Held. et al. (2009)- Imprecise probability assessment.
4. Steffen, W. Rockstrom, J. et al. (2018)-Trajectories of the Earth System.
5. Rietkirk, M. Skiba, V. et al. (2025)- Ambiguity of early warning signals.
6. Scheffer, M. Bascompte, J. et al. (2009)-Early warning signals for critical transitions.

Question: what perspectives might be missing?

Feel free to ask questions at any point during the presentation!

Bias in AI: Types (1/2)

Bias Type	How It Shows Up in the Example	Why It Happens
Western bias	6 of the 6 articles come from scholars based in Western academic institutions.	Training data over-represents English-language, high-impact Western journals.
Prestige bias	Classic, highly cited papers dominate; recent community-based or non-English studies are missing. “Rich get richer.”	The model “learns” that frequently quoted articles are more “salient.”

Bias in AI: Types (2/2)

Bias Type	How It Shows Up in the Example	Why It Happens
Bias in study type	The articles all focus on large-scale evidence and consequences of the climate tipping point, neglecting local effects that are also important to understanding climate change.	Quantitative experimental papers appear more often online than small-sample or local-level work.
Gender / editor influence	All of the cited first authors are male despite contributions by female scholars in climate research.	Historical citation patterns in the corpus perpetuate disparities.

Bias in Results: Claude 4.6 Sonnet

Prompt: Write a sentence about the most important practitioners of public science in the last 100 years. (Claude Sonnet 4.6, June 10, 2026)

Response: Public science communication has been shaped by towering figures like Carl Sagan, whose *Cosmos* television series and prolific writing brought cosmology and critical thinking to millions, and David Attenborough, whose decades of natural history documentaries have made the wonders — and fragility — of life on Earth viscerally real to global audiences.

What bias might be present in this paragraph?

Countering Bias

Feel free to ask questions at any point during the presentation!

Countering Bias: Female Scientists

Prompt: Write two sentences about significant women practitioners of public science. (Claude Sonnet 4.6, June 10, 2026)

Response: Women have made profound contributions to public science communication, with figures like Rachel Carson, whose 1962 book *Silent Spring* almost single-handedly launched the modern environmental movement, and Jane Goodall, whose decades of fieldwork and tireless global advocacy transformed both our understanding of chimpanzees and public attitudes toward conservation.

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Countering Bias: Global South Scientists

Prompt: Please write two sentences about the greatest practitioners of public science from the Global South. (Claude Sonnet 4.6, June 10, 2026)

Response: The Global South has produced remarkable science communicators, including Vandana Shiva, the Indian physicist and environmental activist whose prolific writing and advocacy on biodiversity, seed sovereignty, and the dangers of industrial agriculture have reached audiences worldwide, and Fabiola Gianotti, the Italian-Argentine physicist who as Director-General of CERN has become one of the most visible faces of particle physics globally.

Countering Bias: Discussion

- Try fact-checking the last two paragraphs. Are there any mistakes? Why are mistakes here especially significant?
- What groups of scientists are still excluded from these summaries?
- What other biases may they contain?
- What are biases that we cannot counter by changing prompts?

Countering Bias: Prompts and Feedback

- Bias can be partly countered through careful prompt design and feedback (Dwivedi et al., 2023). However, these methods do not remove bias intrinsic in the model (Shin et al., 2024)
- Practices for countering bias
  - Identify gaps or inconsistencies in generative AI responses.
  - Use additional inquiry and prompt revision to help fill in gaps.
  - Double check responses with other sources.

Personal Impacts

Feel free to ask questions at any point during the presentation!

Critical Thinking

- Recent studies show that the impact of AI on critical thinking depends on how it is used.
- In instances where AI is being used to assist in accessing and analyzing information and in self-directed learning it may support critical thinking (Melisa et al. 2025; Deep and Chen., 2025)
- However, critical reflection and evaluation and writing skills may be hampered by over-reliance on AI (Melisa et al., 2025; Deep and Chen., 2025)
- High self-confidence also fosters critical thinking, while high confidence in AI may result in less critical thinking (Lee et al., 2025)

Language and Expression

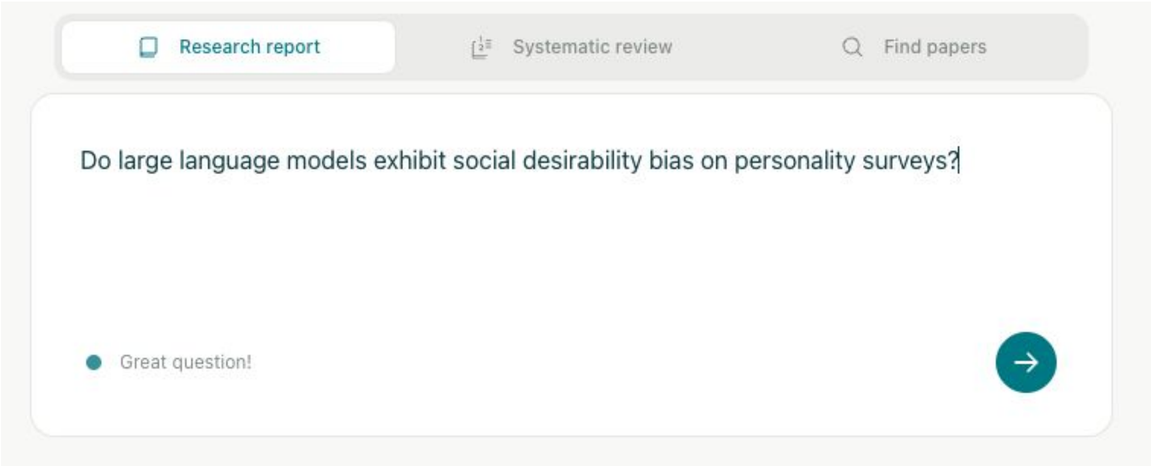
- AI models also vary significantly in their ability to reflect cultural and linguistic nuance depending on the digital prevalence of the language (Shan et al., 2025).
- While AI improves the efficiency of translation it is not as effective for cultural references (Charles-Kenechi, 2024).
- Use of AI assistance in writing can can hamper the expression of cultural nuance and "reinforce Western cultural hegemony" (Agarwal et al., 2025).

Other AI-Powered Literature Review Tools

Feel free to ask questions at any point during the presentation!

Elicit

- Elicit uses AI to generate literature reports
- Provides feedback on research question



Research question from Salecha et al., (2024)

Elicit: Results (Research Report)

Social Desirability in Language Models | Research report View only | Create alert | Share

SEPTEMBER 23, 2025

Do large language models exhibit social desirability bias on personality surveys?

Large language models exhibit social desirability bias on personality surveys, with models like GPT-4 and Llama 3 shifting their responses by over one human standard deviation toward more socially favorable traits such as increased extraversion and reduced neuroticism.

ABSTRACT

Large language models exhibit responses that shift toward socially desirable traits when completing personality questionnaires. Salecha et al. (2024a, 2024b) report that models such as GPT-4 and Llama 3 change their survey responses by 1.20 and 0.98 human standard deviations, respectively, toward increased extraversion and reduced neuroticism when the context suggests evaluation. Reverse-coded items reduced but did not eliminate these shifts, ruling out simple acquiescence bias. Lee et al. (2024a) found that adding commitment statements raised Social Desirability Response index scores from a mean of 4.60 to 5.49 while slightly lowering civic engagement scores. Other studies indicate that bias magnitudes are influenced by model sophistication and even synthetic demographic cues, with some assessments (including the Big Five, Short Dark Triad, and HEXACO instruments) revealing systematic trends toward socially preferred responses.

METHODS

We analyzed 10 sources from an initial pool of 50, using 7 screening criteria. Each paper was reviewed for 5 key aspects that mattered most to the research question.

More on methods

Report

Status

- Gather sources — 50 sources found — Details
- Screen sources — 10 sources included — Details
- Extract data — 50 data points extracted — Details
- Generate report — Save PDF

Chat

Ask anything about the results

Study	Study Focus	Large Language Model (LLM) Models	Experimental Design	Bias Detection
Salecha et al., 2024a	Social desirability bias in large language models on Big Five personality surveys	GPT-4 (0613), GPT-3.5, Claude 3 Opus/Haiku, PaLM-2, Llama 3 70B Instruct, Llama 2 70B Chat	Systematic variation of question number, context, coding, and temperature; multiple models	Score shifts tow as number of qu effect sizes in h deviations; reve paraphrasing, r
Salecha et al., 2024b	Social desirability bias in large language models on Big Five personality surveys	GPT-4 (0613), GPT-3.5, Claude 3 Opus/Haiku, PaLM-2, Llama 3 70B Instruct, Llama 2 70B Chat	Systematic variation of question number, coding, paraphrasing, temperature; multiple models	Score shifts tow as number of qu effect sizes in h deviations; reve paraphrasing, r
Lee et al., 2024	Social desirability response (SDR) bias in GPT-4 survey simulations	GPT-4 (gpt-4-1106-preview)	Within-subjects: presence/absence of commitment statement; cross-cultural personas	SDR index (sum responses), civi index; comparis commitment sta

Feel free to ask questions at any point during the presentation!

Elicit: Results (Find Papers)

Social Desirability Bias in Language Models

Share | Create alert

Do large language models exhibit social desirability bias on personality surveys?

Summary of top 4 sources | Copy

Recent research reveals that large language models (LLMs) exhibit significant social desirability bias when completing personality surveys. Salecha et al. (2024) demonstrated that LLMs systematically skew their responses toward socially desirable traits (increased extraversion, decreased neuroticism) when they infer they are being evaluated on personality dimensions. This bias was observed across multiple models including GPT-4, Claude 3, Llama 3, and PaLM-2, with effect sizes reaching 1.20 standard deviations for GPT-4 and 0.98 for Llama 3. The bias persisted despite question randomization and paraphrasing, and reverse-coding reduced but did not eliminate the effect, ruling out simple acquiescence bias. Lee et al. (2024) found mixed evidence for social desirability response bias in GPT-4 when simulating responses from different societies, with commitment statements increasing bias indices but reducing civic engagement scores. Geng et al. (2024) identified fundamental cultural, age, and gender biases in LLM survey responses, emphasizing the importance of analyzing prompt robustness before using LLMs to simulate social surveys.

Sort: Most relevant | Filters | Export as | UPGRADE

Paper	Abstract summary
Large language models display human-like social desirability biases in Big Five personality surveys Aadesh Salecha +5 PNAS Nexus 2024 · 25 citations DOI	Large language models exhibit human-lik desirability biases when completing Big F surveys.
Large Language Models Show Human-like Social Desirability Biases in Survey Responses Aadesh Salecha +5 arXiv.org 2024 · 12 citations DOI	Large language models exhibit human-lik desirability bias when responding to pers
Exploring Social Desirability Response Bias in Large Language Models: Evidence from GPT-4 Simulations Sanguk Lee +4 arXiv.org	Large language models like GPT-4 may e desirability response bias, but the evider

Manage Columns

Search or create a column

Describe what kind of data you want to extract

e.g. Limitations, Survival time

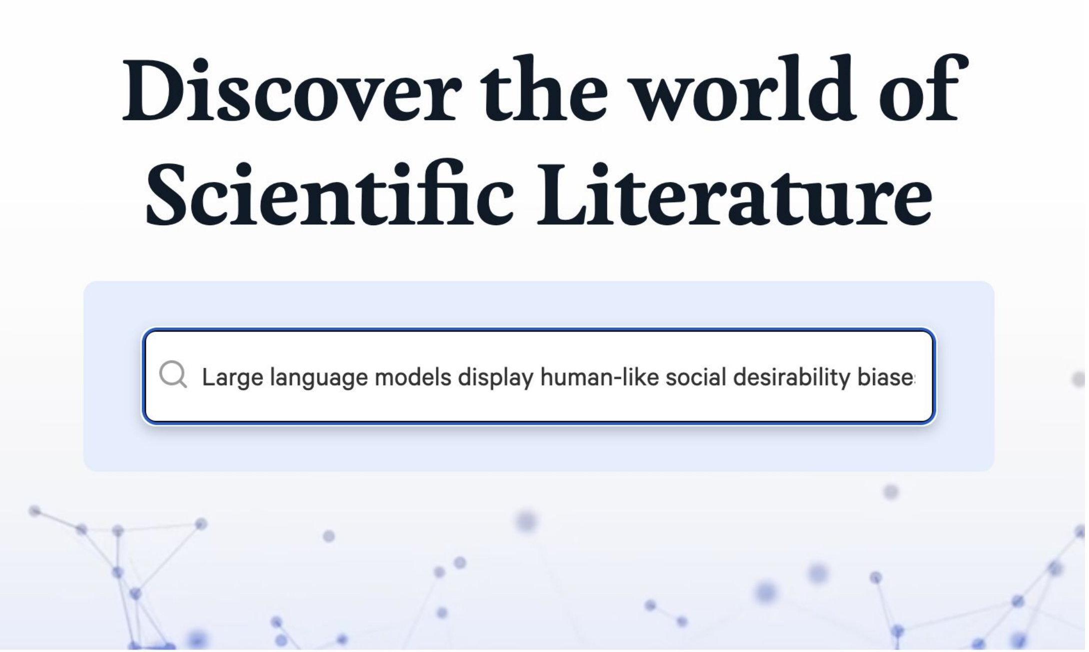
ADD COLUMNS

- Summary
- Main findings
- Methodology
- Intervention
- Outcome measured
- Limitations

Show more

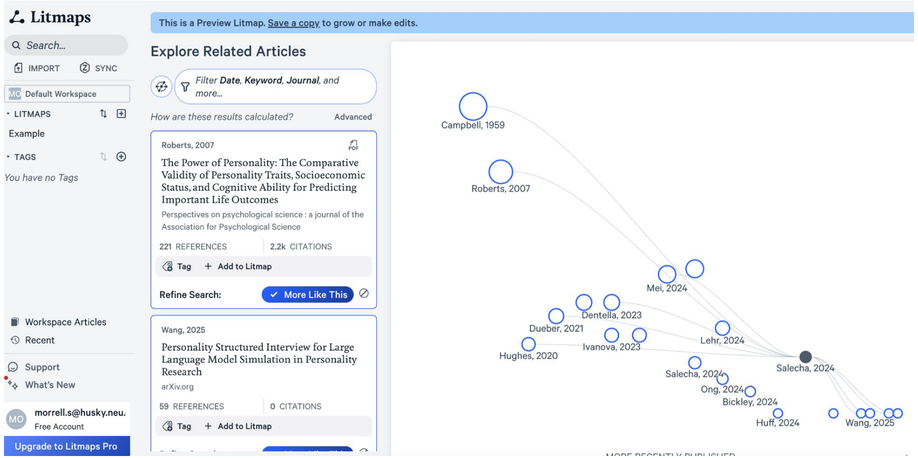
Litmaps

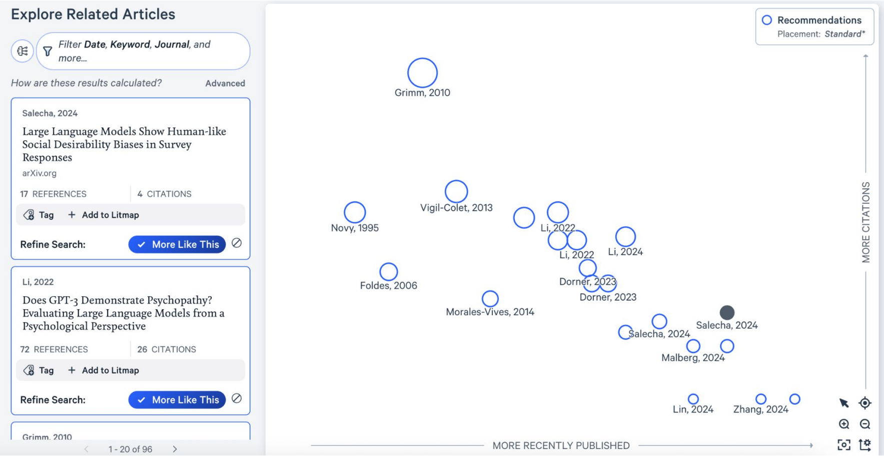
- Litmaps maps relevant literature based on citations, authors, or textual similarity
- Uses AI to find articles with similar text



Feel free to ask questions at any point during the presentation!

Litmaps: Results





Mapping by shared references and citations

Mapping by textual similarity

Other tools

- Scite.ai, finding and summarizing papers, provides context for number of citations by explaining how the paper was used in each of its citations (licensed by Northeastern). Handout here.
- Scopus, develops search queries and creates literature summaries in text or image format. (licensed by Northeastern). Handout here.
- docAnalyzer.ai, document analysis.
- Consensus, finds papers and evaluates the extent to which their results agree.
- Research Rabbit, finds and organizes papers.

Conclusion

Main Points

- AIs, and the data used to train them, are biased
- You can design inputs to help counteract biases
- You should always confirm AI output
- There are multiple AI tools that may be useful at different points in your work

Reflection

1. Have your perspectives changed on AI after this class? If so, how?
2. How might you use the AI tools differently?

Feel free to ask questions at any point during the presentation!

For Further Exploration

DITI Handouts:

Copyright and fair use handout

Data Ethics handout

Data Privacy handout

Northeastern University Resources:

Northeastern Policy on the Use of AI

Generative AI in Teaching and Learning

ChatGPT to Support Reading Development and Critical Thinking

Standards for the Use of Artificial Intelligence in Research

Thank you!

—**Developed by:** Zhen Guo, Sara Morrell, Sean Rogers, Claire Tratnyek, Vaishali Kushwaha, Yana Mommadova, Colleen Nugent, Tieanna Graphenreed, Javier Rosario, Ana Abraham & Chris McNulty, Emily Sandercock, Sayyara Poliferno

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- Schedule an appointment with us!
- If you have any questions, contact us at: nulab.info@gmail.com
- We'd love your feedback! Please fill out a short survey here: https://bit.ly/diti-feedback

Feel free to ask questions at any point during the presentation!

References (1/3)

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NULab Faculty Research on Data, Algorithms, & AI

- "John Wihbey and Christo Wilson on Tiktok Data Espionage Concerns"
- "Alan Mislove Co-Authors Research on Discriminatory Ad Algorithms"
- "John Wihbey Weighs In On AI's Potential to Impact the 2024 Presidential Election"
- "Tina Eliassi-Rad Co-Creates New AI Model that Predicts Human Lifespan"
- "Nabeel Gillani Interviewed by Tech Talk Podcast on AI and Education"

(Cont.) NULab Faculty Research on Data, Algorithms, & AI

- "John Wihbey Participates in a Panel on Content Moderation"
- "John Wihbey Comments on Google's New 'AI Overview'"
- "John Wihbey on the Politics of AI"
- "John Wihbey Interviewed on AI and Epistemic Risk"
- "Malik Haddad on the Regulation of AI"

How to use AI responsibly EVERY time

- "**Evaluate** the initial output to see if it meets the intended purpose and your needs."
- "**Verify** facts, figures, quotes, and data using reliable sources to ensure there are no hallucinations [or confabulations] or bias."
- "**Engage** in every conversation with the GenAI chatbot, providing critical feedback and oversight to improve the AI's output."
- "**Revise** the results to reflect your unique needs, style, and/or tone. AI output is a great starting point, but shouldn't be a final product."
- "**You** are ultimately responsible for everything you create with AI. Always be **transparent** about if and how you used AI."
- Source from [AI for Education]

Feel free to ask questions at any point during the presentation!

AI Plagiarism Checkers

Feel free to ask questions at any point during the presentation!

Plagiarism Checkers: Summary

- Some companies sell tools that claim to identify whether text is AI-generated or human-generated.
- They do this by calculating how statistically predictable each word in the text is.
- AI-checkers can be **wrong**: If a text consistently uses predictable words it is more likely to be labeled as AI generated.

Feel free to ask questions at any point during the presentation!

Plagiarism Checkers: Biases

- These tools have the potential for false positives (identifying human texts as AI).
- False positives are especially likely for texts by writers for whom English is not their first language or for writers who have autism, ADHD, dyslexia, or related neurodivergence.
- To make things worse, writers can often *reduce* their "AI score" by using an AI to reword their essays.

Feel free to ask questions at any point during the presentation!